Parity norms for a smaller Diophantine consistency equation

Thirteen integer unknowns, degree 1,392, and 211 binary arithmetic operations

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Abstract

We construct an explicit polynomial in thirteen unrestricted integer unknowns. Its exact total degree is 1392, and an attached straight-line program evaluates it in 211 binary additions, subtractions, and multiplications. Its absence of integer zeros is equivalent to Con(ZFC), by a new arithmetic reduction composed with the retained ProveIt native/Pell/refutation calibration.

The main device combines five square conditions with fixed powers of two. Odd squareclass coefficients cannot cancel when their powers of two are distinct. This replaces large real separating weights by a parity argument. The main Pell congruence is encoded in the center of the same norm, removing its separate witness. A polynomial change of the remaining norm coordinate then lowers degree from 1888 to 1392.

The contribution is an ordinary proof, with exact algebraic and computational certificates. It does not supply a new end-to-end proof-assistant certificate for the inherited native compiler. It neither assumes ZFC consistency nor asserts global optimality or worldwide priority.

1 Result, comparison, and evidence boundary

There is no single order called *simplest* on Diophantine equations. Unknown count, degree, arithmetic circuit size, coefficient data, and proof complexity are separate measures. The principal new point is

$13\mathbb{Z}$-unknowns / deg 1392 / 211 binary operations.

The unshifted version costs 205 operations at degree 1888. Both polynomials use the same six literal fixed integers. Their coefficient payloads total 615,883 uncompressed bytes; they are not hidden existential parameters.

The main degree improves the placed thirteen-integer comparison by a factor of 10,461,744/1392, about 7516. Its operation count falls from 519 to 211, about 59.3%. These statements concern these measured coordinates, not coefficient size or every construction in the repository. In particular, nine-positive and ten-integer equations remain available. A ten-positive incoming equation also exists. We do not claim a new record for the absolute number of unknowns.

The repository was inspected at the snapshot printed on the title page. The source directories and all eleven incoming entries were inventoried. The new parent is the seventeen-integer degree-88 packet in `con_zfc_pell_congruences_2026-10-23.zip`, including its coherent source-34 coefficient tuple

Construction	Integer unknowns	Exact degree	Operations
Placed source 67, Jones reduction	13	10,461,744	519
New five-root norm, unshifted	13	1,888	205
New five-root norm, shifted	**13**	**1,392**	**211**
New four-root companion	14	862	150
New three-root companion	15	204	128
New centered two-root companion	16	120	123
Incoming Pell-congruence parent	17	88	not used here

Table 1: Like-domain comparisons. The new companions trade more unknowns for lower degree. Counts charge every live binary $+, -, \times$, including squares and multiplication by fixed numerals. Reuse and fixed leaves are free.

[1, 2, 5]. The archive contains the complete parent package, its proof dependencies, actual fixed numerals, and independent audits. New formulas are given in this report and in literal JSON arithmetic programs.

Logical status. The new square-combination, denominator, coordinate, and degree arguments below are proved for unbounded integer inputs. The native interpretation of the inherited source is imported with its ordinary mathematical proof. The resulting theorem is intended as a derivation formalizable in ZFC. The existing project's incomplete internal-model/compiler formalization is not completed by finite tests or by this report. This distinction matters: an ordinary proof composed with an existing theorem, a checked arithmetic identity, and a checked encoded ZFC derivation are different claims.

2 The exact thirteen-variable polynomial

2.1 Variables and constants

There are exactly thirteen independently quantified variables:

$$j,\ C,\ z,\ h,\ r,\ v,\ w,\ T,\ c_0,\ s,\ p_1,\ p_2,\ t \in \mathbb{Z}.$$

Every other symbol in this section is a polynomial abbreviation. In the machine-readable files the names C, v, w, s, t appear as `C_native`, `v_scale`, `w3`, `pair_root`, and `t_parity`.

There are six fixed positive integer coefficients

$$\beta, \quad K_0, \quad M_C, \quad M_F, \quad A_F, \quad A_C.$$

They are the unchanged source-34 tuple and the source-71 determinant scales:

$$\beta = 2^{1064921} - 1, \qquad A_F = 2^{1113}, \qquad A_C = 2^{666104}.$$

The other three are fixed finite sums over the supplied literal K215 tile table, specified by the attached source-34 builder. Section 9 explains their complete binding. In particular M_F is already shifted: $M_F = M_F^{\text{native}} + \beta$. It must not be shifted a second time.

2.2 Polynomial registers

Define, in the displayed order,

$$\begin{aligned}
&J = 6j + 1, && q = \beta J + 1, && W = 4(3w + 1),\\
&c = 2c_0 + 1, && i = 2r + 1, && F = (K_0 + W)C - z(q - 1),\\
&X = qW, && Y = (2v + 1)q^3, && E = XY,\\
&a = E + Y, && H = 4a + 3, && \Delta = a^2 + H = (a + 2)^2 - 1,\\
&R = (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J,\\
&k = R + 1 + hE, && \eta = c - kY, && \zeta = k - \eta.
\end{aligned} \tag{1}$$

The normalized strong and auxiliary registers are

$$\begin{aligned}
&F_2 = 1 + \Delta i^2 c^4, && I = \Delta i c^2,\\
&V = F_2(cT - R) - c, && Q = I^2, && \delta = Q - 1,\\
&r_a = QV^2 - 1, && C_a = \delta r_a.
\end{aligned} \tag{2}$$

The square radicands and the common-sign guard are

$$\begin{aligned}
&A_0 = 1 + XY^2(XY^2 + 1)k^2,\\
&G = (A_F F)(A_C C) - (q - A_F F - A_C C)^2, && Z_\eta = \eta\zeta,\\
&b_g = GZ_\eta - (s^2 - G - Z_\eta + 1)^2 - 1 - p_1^2 - p_2^2,\\
&B_g = 4b_g + 1, && D_0 = 1 + \Delta c^2, && b_0 = X + ac.
\end{aligned} \tag{3}$$

Finally set

$$\begin{aligned}
&Z_0 = H(2t + 1) + b_0, && L = 8H(2Q - 1)V, && Z = Z_0 + L,\\
&d_0 = H^2 A_0, && d_1 = D_0, && d_2 = 16H^2 B_g,\\
&d_3 = 16H^2 F_2, && d_4 = 256H^2 C_a.
\end{aligned} \tag{4}$$

2.3 A polynomial-only norm definition

For fixed m, let

$$\mathcal{N}_m(Z; d_0, \dots, d_{m-1}) = \prod_{\varepsilon \in \{-1,1\}^m} \left(Z + \sum_{\ell=0}^{m-1} \varepsilon_\ell \sqrt{d_\ell} \right).$$

This displayed product explains the proof, but it is an ordinary polynomial with integer coefficients: changing any formal square-root sign permutes its factors, so only even powers of every formal root remain. Replacing each root square by d_ℓ gives the polynomial. All signs and all limits are fixed finite sets.

For a definition without any radical notation at all, use the following four-root evaluator. For inputs x, a, b, c, d, put

$$\begin{aligned} p &= ab, & q_* &= cd, & u &= a+b, \quad v_* = c+d, \\ \sigma &= x^2 - u - v_*, & K_* &= \sigma^2 - 4uv_* + 4(p+q_*), & & \\ L_* &= -4(\sigma + 2v_*), & M_* &= -4(\sigma + 2u), & N_* &= -8, \\ E_* &= K_*^2 + pL_*^2 - q_*(M_*^2 + pN_*^2), & & & & \\ O_* &= 2K_*L_* - 2q_*M_*N_*. & & & & \end{aligned}$$

Then

$$\mathcal{N}_4(x; a, b, c, d) = E_*^2 - pO_*^2. \tag{N4}$$

To obtain $\mathcal{N}_5$, perform these same additions and multiplications on ordered pairs with the rules

$$(A, B) + (C, D) = (A + C, B + D), \qquad (A, B)(C, D) = (AC + eBD,\ AD + BC).$$

Evaluate (N4) at $x = (Z, 1)$, the four radicands $(d_0, 0), \ldots, (d_3, 0)$, and $e = d_4$. If its result is the pair (U_*, V_*), define

$$\boxed{\mathcal{N}_5(Z; d_0, \ldots, d_4) = U_*^2 - d_4 V_*^2.} \tag{N5}$$

These are finite polynomial definitions over $\mathbb{Z}$. The pairs are arithmetic registers, not Diophantine unknowns. The evaluator never computes a square root, a quotient, or a variable exponent.

The proposed equation is simply

$$\boxed{P_{13}(j, C, z, h, r, v, w, T, c_0, s, p_1, p_2, t) := \mathcal{N}_5(Z; d_0, d_1, d_2, d_3, d_4) = 0.} \tag{6}$$

There is no additional equation, positivity requirement, or hidden quantified coordinate. The unshifted companion replaces Z by Z_0.

3 Why parity makes the norm faithful

A norm equation alone generally does not force its constituent radicands to be squares. For example $\sqrt{5} - \sqrt{5} + 1 = 1$; an unweighted three-root norm can have an integer root even when two of its radicands are nonsquares. The following restricted lemma is the new useful mechanism.

Lemma 3.1 (Parity-separated squareclasses)**.** *Let $m \geq 2$, let $a_0, \ldots, a_{m-2}$ be odd integers, and let a_{m-1} be an arbitrary integer. A rational value of a signed sum*

$$\sum_{\ell=0}^{m-1} 2^\ell \varepsilon_\ell \sqrt{a_\ell}, \qquad \varepsilon_\ell \in \{-1, 1\},$$

is possible only if every a_ℓ is an integer square. Conversely, if they are all squares, its signed sums are integers, and all are odd.

Proof. Each nonzero integer radicand has a unique expression db^2, with d a signed squarefree integer and b a positive integer. For an odd radicand, both d and b are odd. Group the signed radicals by d. Distinct signed squareclasses are linearly independent over $\mathbb{Q}$ in the multiquadratic field they generate. This follows, for example, by projecting onto the characters of its finite Galois group. Hence the coefficient of every nontrivial squareclass must vanish in a rational sum.

But in any class containing one of the first $m-1$ terms, its term with the smallest power of two has an odd coefficient after dividing by that power. All later terms have even coefficients. Their signed sum cannot vanish. A class containing only the last nonzero term cannot vanish either. Zero last radicands can be omitted. Therefore no nontrivial squareclass occurs. Negative radicands would have a nontrivial signed class too, so they are also excluded. Rational square roots of integers are integers. The first root is odd and all subsequent weighted roots are even, proving the converse parity statement. □

No order or positivity of the radicands is assumed in this lemma. It remains valid before the guard is known to be positive. The weights $1, 2, 4, \dots$ are constants, independent of the sizes of the radicands. For the actual five radicands,

$$d_0 = H^2 A_0, \quad d_1 = 4(D_0/4), \quad d_2 = 16H^2 B_g, \quad d_3 = 64(H^2 F_2/4), \quad d_4 = 256H^2 C_a.$$

The first four unweighted radicands are odd integers, as proved next. The divisions by four in this sentence are proof notation only; the literal polynomial uses D_0 and $16H^2F_2$ directly.

4 Unconditional arithmetic facts and denominator recovery

Lemma 4.1 (Ambient congruences)**.** *For every assignment of the thirteen integer inputs,*

$$q \equiv 0 \pmod 2, \quad X \equiv Y \equiv a \equiv 0 \pmod 8, \quad \Delta \equiv 3 \pmod 8, \quad H \text{ odd},$$

$$A_0 \text{ odd}, \qquad B_g \equiv 1 \pmod 4, \qquad D_0 \equiv F_2 \equiv 4 \pmod 8.$$

Moreover $H \neq 0$, $\Delta > 1$, $I \neq 0$, and $Q > 1$.

Proof. The coefficient β and $J = 6j + 1$ are odd, so q is even. The nonzero odd integer J has absolute value at least one, and the fixed β is very large; in particular $q \neq 0$ and $|q| > 4$. The integer $W = 4(3w+1)$ is a nonzero multiple of four. Thus $X = qW$ is a multiple of eight, and the odd multiple $Y = (2v+1)q^3$ is a nonzero multiple of eight. This gives $a = Y(X+1) \equiv 0 \pmod 8$. Since $|Y| \geq |q|^3$ and $X + 1 \neq 0$, $|a| \geq |q|^3 > 4$; hence $(a+2)^2 - 1 > 1$. Now c, i are odd and nonzero, so $I = \Delta i c^2 \neq 0$ and $|I| > 1$. The remaining congruences follow by reducing the displayed definitions modulo eight, using $c^2 \equiv i^2 \equiv c^4 \equiv 1 \pmod 8$. □

The rational auxiliary radicand r_a/δ is multiplied by δ^2, giving $C_a = \delta r_a$. Square recovery gives a rational auxiliary root; the following elementary lemma proves that it is integral.

Lemma 4.2 (Rational adjacent-Pell descent)**.** *Let $Q \geq 2$ and $V \in \mathbb{Z}$. If $y \in \mathbb{Q}$ satisfies*

$$QV^2 - (Q-1)y^2 = 1,$$

then $y \in \mathbb{Z}$.

Proof. Write $y = a/b$ in lowest terms, $b > 0$. Then $b^2 \mid Q - 1$, so $(Q-1)y$ is an integer. Take $V > 0, y \geq 0$ by signs. The equation excludes $V = 0$, and $V = 1$ gives $y = 1$. For $V \geq 2$, $y > V$. Define

$$V' = (2Q-1)V - 2(Q-1)y, \qquad y' = (2Q-1)y - 2QV.$$

The first quantity is integral. Both satisfy the same norm equation. Also $V' < V$, since $y > V$. To see $V' > 0$, observe

$$(2Q-1)^2V^2 - 4(Q-1)^2y^2 = V^2 + 4(Q-1) > 0.$$

Consequently repeated replacement by $(V', |y'|)$ strictly decreases the positive integer V until it reaches one. The transformation has an inverse with integer coefficients. At the terminal pair $y = 1$; reversing every step proves that the original rational y was integral. □

Remark 4.3. No squarefreeness of $Q - 1$ is used. Merely clearing a denominator without this argument would leave a proof gap.

5 The imported source and the new reconstruction

5.1 The parent interface

The retained seventeen-integer source uses the same β, K_0, M_C, M_F, the same determinant scales, and the same polynomial transport

$$F = (K_0 + W)C - z(q - 1).$$

Its free main root is represented by $D = X + ac + H\gamma$. It has odd scale parameters in Y and in an ordinary strong coordinate $I_{\text{old}} = (2r_{\text{old}} + 1)c^2$. Besides its monic registers and transport, its relevant equations can be written

$$\begin{aligned} \tau^2 &= A_0, & D^2 &= 1 + \Delta c^2, \\ I_{\text{old}}^2 &= \Delta(f^2 - 1), & I_{\text{old}}^2 V^2 - (I_{\text{old}}^2 - 1)y^2 &= 1, \qquad (7) \\ V &= fcT_{\text{old}} - f^2 R - c, \end{aligned}$$

$$GZ_\eta - (s^2 - G - Z_\eta + 1)^2 = 1 + p_1^2 + p_2^2 + p_3(p_3 + 1). \qquad (8)$$

The packet centers its first and auxiliary roots; the inverse changes are

$$\tau_{\text{packet}} = 2\tau - 2P + k, \qquad y_{\text{gap}} = y - V, \qquad P = k + XY^2 k.$$

These are integral polynomial expressions in the restored roots.

The parent report proves that its integer zeros correspond to finite refutations of the specified ZFC calculus. Its main ingredients are the nearest-square joint-positivity guard, determinant sign recovery, signed Pell rank recovery, residue orientation of X, a CRT compression of two auxiliary congruences, and the fixed native-window compiler [2, 3, 6, 7]. The new construction restores this exact interface rather than silently switching to another coefficient tuple.

5.2 Every new zero restores an integer parent witness

Theorem 5.1 (Soundness of the new arithmetic reduction)**.** *Every integer zero of* (6) *gives an integer zero of the retained seventeen-integer source.*

Proof. Let $x = 2t + 1 + 8(2Q - 1)V$, so $Z = Hx + b_0$ and x is odd. The norm factorization and Lemmas 3.1–4.1 force all five radicands to be integer squares. Dividing their integer roots by the displayed square factors, using rational-square integrality, gives signed integers

$$\tau,\ D,\ d,\ f,\ z_a$$

such that

$$\tau^2 = A_0, \quad D^2 = D_0, \quad d^2 = B_g, \quad f^2 = F_2, \quad z_a^2 = C_a.$$

Signs can be chosen to make the vanishing factor read

$$Hx + b_0 = H\tau + D + 4Hd + 4Hf + 16Hz_a. \tag{9}$$

In particular

$$\gamma = \frac{D - b_0}{H} = x - \tau - 4d - 4f - 16z_a$$

is an integer. Thus the actual main congruence is restored, not just the main square test. The root d is odd, so $p_3 = (d-1)/2$ is an integer. Its square identity gives exactly (8).

Set $y = z_a/\delta$. Since $z_a^2 = \delta(QV^2 - 1)$,

$$QV^2 - (Q-1)y^2 = 1.$$

Lemma 4.2 makes y integral. For the ordinary strong source put

$$r_{\text{old}} = \frac{\Delta i - 1}{2}, \qquad T_{\text{old}} = fT.$$

These are integers, since Δi is odd. Then

$$(2r_{\text{old}} + 1)c^2 = I, \qquad I^2 = \Delta(F_2 - 1) = \Delta(f^2 - 1),$$

and

$$fcT_{\text{old}} - f^2R - c = f^2(cT - R) - c = V.$$

The old affine J-coordinate is $2j$; its old c coordinate is $2c_0 + 1$. The centered-root inverses above complete all seventeen coordinates. Every parent residual is now zero.

For completeness of the sign discussion, (8) has a strictly positive right side. The inherited nearest-square guard therefore gives $G > 0$ and $\eta\zeta > 0$, and the determinant guard recovers the required common sign of $F, C, q - F - C$. Those conclusions are obtained before invoking the source's signed rank and native semantics. □

The proof never assumes that arbitrary inputs are positive, that the radicands are positive before extraction, or that a canceled denominator is nonzero without a reason.

6 Completeness without an extra congruence promise

The normalized chart is a genuine restriction of the set of parent witnesses. Thus a proof that every parent zero has a polynomial inverse would be incorrect. What is needed is that every hypothetical finite refutation supplies at least one canonical witness in this chart. We prove that statement explicitly.

The retained source-34 canonical family has

$$q = 2^\lambda, \quad \lambda = 1064921N \text{ odd}, \quad R \equiv 3 \pmod 4, \quad R > 3q,$$

$$X = 2^R, \quad Y/q^3 \text{ odd}, \quad A = a + 2 \text{ even}, \quad c = \psi_A(R) \text{ odd}.$$

Here $\chi_A(n) + \psi_A(n)\sqrt{A^2 - 1} = (A + \sqrt{A^2 - 1})^n$. It also supplies both the determinant guard and the source residue conditions simultaneously. In particular $J \equiv 1 \pmod 6$, and $W = 4(3w + 1)$. These are documented properties of the actual same source-34 tuple, not a borrowed property of the smaller compact tuple.

Lemma 6.1 (An odd normalized strong witness)**.** *For this canonical $c = \psi_A(R)$, there are an odd positive integer i and an integer $f > 0$ satisfying $f^2 - \Delta i^2 c^4 = 1$.*

Proof. Choose $m = Rc$, $f = \chi_A(m)$, and $z_* = \psi_A(m)$. Let $d = \chi_A(R)$. The composition identity gives

$$z_* = c\,\psi_d(c).$$

Since $d^2 - 1 = \Delta c^2$, the binomial formula for the Pell sequence gives

$$\psi_d(c) \equiv c\,d^{c-1} \pmod{c^2}.$$

Consequently $c^2 \mid z_*$. As A is even and $m = Rc$ is odd, z_* is odd. Thus $i = z_*/c^2$ is a positive odd integer, and the Pell identity is the required normalized strong equation. □

Lemma 6.2 (The squared-modulus auxiliary choice)**.** *With the preceding canonical witnesses, put $S = \Delta i c^2$. There are positive integers T, V, y such that*

$$V = f^2(cT - R) - c, \qquad S^2V^2 - (S^2 - 1)y^2 = 1.$$

Proof. Write $R = 2v_* + 1$, where v_* is odd, and define

$$V_* = \chi_S(R)/S, \qquad y_* = \psi_S(R).$$

The first expression is an integer polynomial $Q_{v_*}(S^2)$, because $\chi_S(R)$ is an odd polynomial in S. The Pell identity supplies the stated auxiliary equation. The elementary odd-index polynomial identities are

$$Q_{v_*}(0) = (-1)^{v_*}R = -R, \qquad Q_{v_*}(1 - A^2) = (-1)^{v_*}\psi_A(R) = -c.$$

They follow by the binomial formulas for χ and ψ, or by their recurrences. Since $S^2 = \Delta(f^2 - 1) \equiv 1 - A^2 \pmod{f^2}$ and $c \mid S$,

$$V_* \equiv -c \pmod{f^2}, \qquad V_* \equiv -R \pmod{c}.$$

Also $f^2 \equiv 1 \pmod{c}$, hence $\gcd(c, f^2) = 1$. Therefore

$$T = \frac{V_* + c + Rf^2}{cf^2}$$

is a positive integer. It yields the asserted $V = V_*$ and $y = y_*$. This is precisely the squared-modulus reconstruction proved in the retained source-61 lemma [4]; it does not demand an unproved choice of arbitrary congruences. □

Theorem 6.3 (Completeness of the new arithmetic reduction)**.** *A finite refutation of the specified ZFC calculus yields a thirteen-integer zero of (6).*

Proof. Take the simultaneous canonical native witnesses. Set $j = (J - 1)/6$, $c_0 = (c - 1)/2$, and $r = (i - 1)/2$, using Lemma 6.1. Reselect the auxiliary witnesses by Lemma 6.2. Keep the native transport and guard coordinates. The ternary positive representation provides p_1, p_2, p_3, and put $d = 2p_3 + 1$. Let τ, D, γ be the canonical first and main roots and the main quotient.

The congruences give τ odd, D even, and $b_0 = X + ac$ even. Since H is odd, $\gamma = (D - b_0)/H$ is even. Thus

$$x = \tau + \gamma + 4d + 4f + 16\delta y$$

is odd. Define

$$t = \frac{x - 1}{2} - 4(2Q - 1)V \in \mathbb{Z}.$$

Then (9) holds with these signs and $z_a = \delta y$. A factor of the norm vanishes. This constructs an actual integer solution from the hypothetical finite certificate. □

Corollary 6.4 (Consistency equivalence)**.** *Composing Theorems 5.1 and 6.3 with the retained native/Pell/refutation theorem gives, by ordinary mathematics formalizable in ZFC,*

$$\mathrm{ZFC} \vdash \Big(\neg\exists \mathbf{x} \in \mathbb{Z}^{13}\ P_{13}(\mathbf{x}) = 0\Big) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

There is no appeal to the truth of either consistency or inconsistency. All completeness choices are conditional constructions from a finite refutation. The theorem does not conflict with the second incompleteness theorem: it proves an equivalence, not either side of it.

7 The degree reduction and exact degree certificate

The unshifted norm uses Z_0. The shifted version is obtained by the integer polynomial change

$$t_{\mathrm{old}} = t + 4(2Q-1)V.$$

All registers on the right are independent of t, so this is a triangular polynomial bijection with inverse subtraction. Therefore the two thirteen-variable polynomials have equivalent zero sets, with no added number-theoretic condition.

With inputs of degree one and fixed coefficients of degree zero,

register	q	X	Y	E	a	Δ	R	k	I	F_2	V	C_a
deg	1	2	4	6	6	12	5	7	15	18	23	106

The degrees of G, Z_η, B_g are $4, 22, 44$, respectively. Thus the radicand degrees are

$$(\deg d_0, \ldots, \deg d_4) = (46, 14, 56, 30, 118).$$

The carrier Z_0 has degree seven. In the unshifted product, the last radicand strictly dominates; its sixteenth power gives exact degree $16 \cdot 118 = 1888$.

For the shifted norm, pair the two signs of its final radical first. For a signed sum S_4 of the other four radicals, the paired factor is

$$(Z_0 + L - S_4)^2 - d_4.$$

The critical identity is

$$L^2 - d_4 = 64H^2V^2 + 256H^2\delta. \tag{10}$$

It is an exact polynomial identity, since $(2Q-1)^2 - 4Q(Q-1) = 1$. Therefore the paired factor equals

$$(Z_0 - S_4)^2 + 2L(Z_0 - S_4) + 64H^2V^2 + 256H^2\delta.$$

Use weights one-half of each radicand degree for its formal root. The largest root among the four is $\sqrt{d_2}$, of weight 28; $\deg L = 59$. Every paired factor has weight at most 87, with unique highest term $-2L(\pm\sqrt{d_2})$. There are sixteen paired factors. Their product has degree at most $16 \cdot 87 = 1392$, and its highest homogeneous part is

$$\boxed{2^{16}[L]_{59}^{16}[d_2]_{56}^{8}.}$$

This is nonzero for the actual fixed coefficient tuple. The attached independent certificate substitutes specified positive integer slopes times an indeterminate into every input and reduces the actual six coefficient numerals modulo 1,000,000,007. The degree-1392 coefficient is

$$756{,}708{,}762 \neq 0 \pmod{1{,}000{,}000{,}007}.$$

Together with the proved upper bound, this is an exact nonvanishing certificate, not a random estimate of degree. A separate finite-difference calculation reproduces it.

8 The smaller companion equations

The archive retains intermediate constructions because a higher unknown count can buy a much lower degree.

8.1 A two-root merger in sixteen integers

In the seventeen-variable parent, $\tau^2 = A = 4a_4 + 1$ and $\delta y^2 = r_a$, with $\delta \neq 0$. For $x = 2t + 1 = \tau + 2y$, clearing the two-root norm and dividing its fixed content gives

$$N_{16} = \big[\delta(t(t+1) - a_4) - r_a\big]^2 - A\delta r_a.$$

A zero gives

$$\tau = \frac{\delta(x^2 + A) - 4r_a}{2x\delta} \in \mathbb{Q}, \qquad \tau^2 = A.$$

Thus τ is an odd integer and $y = (x - \tau)/2$ is an integer. This particular two-root version needs no adjacent-Pell descent.

For the centered form put $z_0 = XY^2/4$, a polynomial because $X = 4q(3w + 1)$, and substitute the even center $4z_0k$ into x. The resulting numerator is

$$\delta\big[t(t+1) + 4z_0kt + 2z_0k - z_0k^2\big] - r_a.$$

Combining its norm with the three retained parent rows by a sum of squares gives degree 120 in sixteen integers, evaluated in 123 operations.

8.2 Three and four roots

The fifteen-integer companion also packs the guard root $2p_3 + 1$ with weights $1, 2, 4$. The three residual degrees are $14, 14, 176$. For any integer residuals $r_1, \ldots, r_n$,

$$\prod_{\ell=1}^{n}(1 + 4r_\ell) - 1 = 0 \quad \Longleftrightarrow \quad r_1 = \cdots = r_n = 0,$$

because every factor must be an integer unit, and no factor can equal -1 modulo four. This yields degree 204 and 128 operations.

The fourteen-integer companion uses the normalized strong chart, combines four roots, and retains the main congruence witness γ. Its two residual degrees are $14, 848$; (8.2) gives degree 862 and 150 operations. The thirteen-integer construction then absorbs γ into the center and needs no residual finalizer.

These intermediate results explain how the main construction was obtained. They do not make the final polynomial depend on an optimization search.

9 Fixed coefficient specification and provenance

The six values are genuine integer coefficients fixed before any question about halting is asked. Their storage and reproducibility are part of the mathematical artifact.

Coefficient	Bits	Uncompressed bytes
β	1,064,921	133,116
K_0	1,065,043	133,131
M_C	1,064,921	133,116
M_F	1,064,922	133,116
A_F	1,114	140
A_C	666,105	83,264
Total		615,883

The files under `coefficients/payloads/` are minimal unsigned little-endian representations, gzip-compressed. The binding manifest gives both raw and compressed SHA-256 hashes. The source-34 recipe is included, with its literal `k215.tm` and `wang943_tiles.json`. The recipe checks all 430 transitions and all 943 generated tiles before constructing the coefficient sums.

For a compact mathematical summary, it uses radix $\rho = 2^{121}$, block length 8801, and $B = \rho^{8801}$. It builds the fixed selector weights, low mask, high mask, and convolution polynomial D_C from the authenticated tile list, with $D_R = \rho$, then

$$\beta = B - 1, \qquad K_0 = D_C + BD_R, \qquad M_F = M_F^{\text{native}} + B - 1.$$

The two determinant scales are the powers of two already printed above. Every sum and loop in the builder has a fixed finite range; it never waits for K215 to halt. Its exact literal transition hash is

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090`

The parent packet is retained byte for byte. No coefficient from the compact source-44 or original source-31 tuple is substituted for a source-34 port. Equality of radix exponents would not justify such a substitution.

The archive includes the complete incoming parent package as `upstream/pell-congruences-source/`, with its original notices and proofs. This retains the exact source semantics and canonical witness promises required above. A machine-readable source ledger pins the Git commit, archive hash, parent packet, and coefficient binding.

10 Verification and how to reproduce the result

The checks address concrete failure modes.

1. **Universal norm identities.** An independent generator expands the generic four- and five-root polynomials by Newton identities. Their 495 and 20,349 monomials agree exactly with the independently built multiquadratic norm circuits. The compact four-root evaluator is also checked by its biquadratic coefficient identity; lifting it to the final quadratic algebra proves the five-root evaluator.

2. **Literal graph and coefficient checks.** The delivered program contains only live binary $+, -, \times$ nodes. It has 211 operations: 51 additions, 31 subtractions, and 129 multiplications. All fixed coefficients are authenticated. Independent graph evaluation agrees with the separate source packets. The count is an attained upper bound, not a claim that no shorter circuit exists.

3. **Exact degree.** Safe degree bounds, the cancellation-resolving identity (10), a nonzero specialized leading coefficient, and independent finite differences agree on 1392.

4. **Exceptional cases and negative controls.** Tests include signed radicands, zero guard candidates, cleared-denominator degeneracies, omitted oddness, and the false unweighted norm. The proof excludes each relevant false branch for unbounded inputs. The parent two/three-root companions explicitly handle $Q = 0$ when their broader charts allow it; the final normalized thirteen-variable chart has $Q > 1$ on every integer assignment.
5. **Bounded diagnostics.** The independent generic checker examines 119,000 pair assignments, 143,990 triple assignments, 3,360 completeness witnesses, and 7,059 rational-Pell fibers. These are transcription and adversarial diagnostics, not evidence for ZFC consistency or substitutes for the proofs.

From the archive root, the release runner rebuilds the equations, the compact arithmetic programs, and the independent certificates:

```
python3 verify_release.py
```

All new mathematics/code requires only the Python standard library. The report builds with a conventional LaTeX installation:

```
cd report
latexmk -pdf -interaction=nonstopmode -halt-on-error parity-norms.tex
```

The exact paths and commands are recorded in the top-level README. The coefficient recipe and inherited release scripts keep their own original execution conventions; they are not silently relabeled as new proofs.

11 What is new, and what remains open

The useful new idea is a restricted constant-weight square-combination theorem, applied where the source already supplies the needed parities. The method has three parts:

- separate squareclass contributions by powers of two;
- make a norm center carry a congruence as well as a sum of roots;
- remove the leading auxiliary square by an exact midpoint shift.

The parity statement is proved here and appears absent from the inspected ProveIt source and incoming constructions in this role. We do not claim that the general algebra of multiquadratic norms is new, or that this is the first use of parity in Diophantine representation. Literature searches were not a proof of historical priority.

The construction retains K215 as the inherited source of one concrete coefficient tuple and its refutation calibration. Its new compression does not use Jones's nine-variable reduction or Sun's ten-integer reduction. The square-combination lemma is independent of that source and may be reusable for other arithmetic encodings. Sun's ten-integer theorem and the universal-pair literature remain the appropriate broader comparisons [8, 9].

The principal open tasks are precise:

1. Formalize the parity lemma, normalized chart, reconstruction, and midpoint identity inside the existing proof development.
2. Close the inherited native/compiler/internal-model calibration if an end-to-end checked ZFC derivation is desired.

3. Investigate whether another guarded congruence can be moved into the same center. The transport modulus $q-1$ need not be coprime to H; canceling it without a proof would be invalid.
4. Reduce the guard's degree or its witnesses. A claimed two-variable replacement for the ternary positive representation needs its own surjectivity theorem.

Neither absolute minimality nor a smaller-variable universal representation is established by the present result. The delivered thirteen-variable polynomial is a concrete, reproducible improvement at a different point of the repository's size–degree frontier.

References

[1] V. Reshetnikov et al., *ProveIt*, snapshot `cfbf118b4ad9b2c651fe96b571d9415261766328`. https://github.com/VladimirReshetnikov/ProveIt. See `Computability/DiophantineConsistency/` and `Computability/HilbertTenthProblem/`.

[2] ProveIt incoming package, *Pell congruences, shared guards, and monic transport*, `con_zfc_pell_congruences_2026-10-23.zip`. The complete package is retained in this release; the principal parent is `transport/degree88_17.json`.

[3] ProveIt source 34, *A mixed-radix, parity-separated, unrounded coefficient tuple*. The exact coefficient builder and canonical odd-scale proof are retained in the incoming parent.

[4] ProveIt source 61, *Congruence carriers and a divided Pell norm*, `61-congruence-carriers/report/congruence_carriers.tex`, especially the normalized strong and squared-modulus auxiliary lemmas.

[5] ProveIt source 67, *Oriented guards*, `67-oriented-guards/README.md` and its thirteen-integer companion.

[6] ProveIt source 70, *Residue orientation*, the signed source and removal of the separate $X = 2u^2$ witness.

[7] ProveIt source 71, *Determinant guards*, including canonical completeness for the six-port source-34 tuple.

[8] Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594v2 (2026). https://arxiv.org/abs/2609.01594.

[9] J. Bayer, M. David, M. Hassler, Y. Matiyasevich, and D. Schleicher, *Diophantine Equations over $\mathbb{Z}$: Universal Bounds and Parallel Formalization*, arXiv:2506.20909v3 (2026 revision). https://arxiv.org/abs/2506.20909.